Natural History

Alveary

Natural History: Grade 6

GRADE(S): 6

About the Course

Students learn about oceanography, predator-prey relationships, and the impact of pollution as they consider man's stewardship of the natural world. Coordinating afternoon activities are provided in Outdoor Work.

This topic is included in the following course(s): Science: Grade 6

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level and time required are transitional as we gently prepare for the middle grades.

Science: Grade 6
The Big Picture:
To accomplish the goal of supporting a relationship with the Things of the Universe, a Mason science program consists of nature lore, natural history, and general science. Nature immersion, inquiry, community connection, and supportive literature are woven into each of these three parts. Building on the familiarity and foundational knowledge of Form 1, Form 2 learners in grades 4-6 extend the scope of the Things they encounter and begin to notice that science is active in their world. Growing in their self-knowledge at this age, they may notice that they gravitate toward some of these Things more than others. This is a bit like noticing that you have a natural rapport with a particular friend because they seem to understand the way you think and share the same interests. This is wonderful and very typical, but we still want them to get to know more "friends" in science and try different Things, even though one might be their particular interest. Their sense of community is expanding, as well: who are the scientists in their world, and what are they doing? Special care has been taken to build vocabulary and conceptual knowledge "by the way" and to maintain the connection between scientific endeavors and the human experience. In Form 3, students will continue to build on this knowledge of what science is doing in the world when they examine science in its historical context.

Nature lore is timeless knowledge that is passed through a community, much like a grandmother passes on how to make that special bread when the dough just "feels right." Like Mason, we strive to pass on this knowledge primarily through outdoor work. Group nature walks, seasonal readings, and topics in scouting are provided as an Outdoor Work resource in the Quick Links. If desired, literature suggestions to support lore can be found in the Community Read Alouds resource (in Citizenship Grades 4+).

Placement & Combining Tips

About Science: Grade 6
In level 6, learners expand their knowledge of science in their world and continue building personal connections as they discover a variety of topics from oceanography, architecture, and more. Reading level, historical context, and time required are transitional as we gently prepare for the middle grades.

Natural History: Grade 6
For approximately sixth-grade students based on a balance of complexity and reading level. However, learners may be freely combined, or sequence reordered based on needs and preferences.

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
6	Natural History: Grade 6 1 time/week 15 min	The Frog Scientist The Wolves and Moose of Isle Royale: Restoring an Island Ecosystem Tracking Trash

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Grade 6				
Natural History: Grade 6		General Science: Grade 6 Nature Walks & Scouting: Grades 1-8	General Science: Grade 6 Nature Notebook: Grade 6	Labs: Grade 6

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Science: Grade 6
☐ Obtain any supplies indicated on the science or grade-level supply lists.
☐ Download any apps and shortcut any desired links.

Nature Lore:
☐ Bookmark your Outdoor Work Quick Link, so that you have it available on your weekly outing. Outdoor Work is generally flexible for your location and season and can be moved around in the schedule to incorporate or substitute Natural History Club outings.
☐ Print or bookmark grade-specific nature notebook suggestions to support natural history and general science. Notebooking can be done on a walk, during occupations, or as a field trip, as appropriate.

Special Topics & Field Trips

Science: Grade 6
Encourage students to begin choosing their own special topic to observe, study, and research in their field guides. What or who are they curious about or interested in getting to know better? Teachers can choose their own special study, too! Suggested field trips include:

Natural History: Grade 6
☐ Term 1: shore, stream, or reservoir, a marine animal rescue
☐ Term 2: wildlife conservationist
☐ Term 3: a pond (hopefully one where tadpoles may be collected)

Term Prep & Teacher Tips

Science: Grade 6
Object lesson prompts are woven into the lessons, as appropriate, but teachers are encouraged to read corresponding Handbook of Nature Study (HoNS) selections each term and/or any that support special topics chosen by students:

Natural History: Grade 6
☐ Term 1: -
☐ Term 2: -
☐ Term 3: amphibians p.170-187

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

- View Book List Details

Natural History: Grade 6

 The Frog Scientist

 The Wolves and Moose of Isle Royale: Restoring an Island Ecosystem

 Tracking Trash

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

- View Basic Supplies

(No Subject Supplies Assigned)

Quick Links

Click THIS text or scan the QR code for links.

Related Course Quick Links

- Extra Helpings
- Outdoor Work
- Seek app from iNaturalist
- SkyView Lite for iOS
- SkyView Lite for Android
- Foundations (See Section 13: Science)

Natural History: Grade 6

How To Teach

Introduce

• Begin each lesson with learners' existing knowledge. If the book or activity is new or unfamiliar, then look at the title, a picture, or guidance in the lesson plan to help discuss what students think, drawing on previous experience. If continuing or revisiting a topic or activity, then recap.

• Often, the introduction in the lesson plans or the title of the book's section can help learners to draw out the main idea. Use this to support learners, as appropriate.

• Some learners may benefit from using pictures or looking back briefly.

• Allow them time to share any concerns and come alongside, as needed.

Read

• Read or do, as instructed in the lessons, noting any Teacher Tips provided. Learners should always have their journal, notebook, or other paper available in case they need to draw or diagram during the lesson.

• Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.

• If learners do not understand a word or concept, do not worry. Try to show them with a picture or connect the idea to something they have seen in real life. They are learning much by the way and will likely build understanding over the term, the year, and beyond.

Narrate

• Process the ideas of the lesson by retelling events, describing, explaining a concept, etc. Tips about helping learners deal with various types of passages are provided, but teachers can learn more in Charlotte Mason's School Education Chapter 16, especially p.180.

• Learners may use words, pictures, Legos, PowerPoint, etc, to process and convey ideas in their own way.

• Teachers may take turns to model.

Discuss

• Consider together any thoughts, confusion, or concerns about the passage, keeping in mind that oftentimes new concepts are not going to be 'mastered' on the first or even second introduction. Mastery is not necessarily the goal, but curiosity and thinking.

• Questions/topics for further discussion are often provided in the lesson plans (or even lab books) to help. There are no right or wrong answers to these. Alternatively, many of these can be used for composition, depending on the needs of the learner and the instructional goals of the teacher. If teachers want to keep ideas active in the mind, these can also be used at other times to keep the ideas in the working memory.

• Note that learners may need to spend more than the allotted time engaging with or even repeating a lesson before moving on or as reinforcement at a later time. Adjust the pace as needed to feed the learner.

• Notice if there were any dates that they want to keep for their Book of Centuries.

Connect

• Follow any extra links, examine any sidebars in the text, look at pictures, etc., depending on learner needs and interest.

• These can be viewed as alternative ways to engage.

Natural History: Grade 6

Click THIS text or scan the QR code for links.

Term 1

WEEK 1

⬚ **15m Natural History: Grade 6 - Lesson 1**
Ocean Currents

▯ Materials: Tracking Trash

PREP: Read Teacher Tip.

→ INTRO
Look at the picture on the cover and the title. Have you ever noticed trash on the beach or in a stream? This book is about how one scientist uses that trash as an opportunity to learn more about the ocean and the creatures that live within it. Examine the map opposite p.1 and notice the current. Why do you suppose the mapmaker was interested in that current, and what does that have to do with our story?

→ READ, NARRATE, & DISCUSS
p.1-4 "Benjamin Franklin" - "learn from it."

- How do these ocean currents affect the climate? You can study the map on p.20-21 of your NatGeo World Atlas.
- Explain weather versus climate.

★ TEACHER TIP
If students read Tornado Scientist, they might recall that air masses are warmed by the Earth or water beneath them. Movement occurs based on their temperature and density, and this is a driving force behind weather patterns.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 2

⬚ **15m Natural History: Grade 6 - Lesson 2**
An Opportunity

▯ Materials: Tracking Trash

PREP: Read Teacher Tip.

→ RECAP
Recall what this scientist is studying and why. Let's find out how he started tracking trash.

→ READ & NARRATE
p.4-7 "Curt didn't" - "Pacific Ocean."

→ READ, NARRATE, & DISCUSS
p.8-9 "Finding a certain" - "of the spill."

★ TEACHER TIP
This passage has two parts with two different purposes. The first part tells how this experiment began and is more likely to be narrated as a series of events. The second explains latitude and longitude (and why they are important) and is more likely to be narrated as points that describe these concepts. It can be helpful to teach students to identify the purpose of different passages as they prepare to narrate.

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.

WEEK 3

⬚ **15m Natural History: Grade 6 - Lesson 3**
Drift Experiments

▯ Materials: Tracking Trash

→ RECAP
Remind me how Curt started tracking trash. Scientists have known for a long time that ocean motion isn't random. How do you think scientists typically study ocean currents?

→ READ, NARRATE, & DISCUSS
p.11-15 "Throughout this" - "floating sneakers."

• OUTDOOR WORK
Weekly walks and daily outdoor time are vital! Resources in Quick Links.